

Machine Learning Lab Exercise

Linear Regression using the Synthetic Auto MPG Dataset

Objective

Use Linear Regression to predict a car's fuel efficiency (**mpg**) from vehicle characteristics.

Dataset

Use `auto_mpg_synthetic_dataset.csv`.

Features

Feature Description

`mpg` Target variable (Miles Per Gallon)

`cylinders` Number of cylinders

`displacement` Engine displacement

`horsepower` Engine horsepower

`weight` Vehicle weight (lbs)

`acceleration` Acceleration time

`model_year` Model year

`origin` 1=USA, 2=Europe, 3=Japan

`car_name` Car identifier

Exercise 1: Load and Explore

1. Load the dataset using Pandas.
2. Display the first five rows.
3. Display the dataset shape.
4. Display `info()`.
5. Display `describe()`.

Questions

1. How many observations are in the dataset?
 2. Which column is the target variable?
 3. Which column contains text?
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Exercise 2: Missing Values

1. Check for missing values.
2. If any exist, remove them.
3. Display the new dataset shape.

Questions

1. Which columns contain missing values?
 2. Why should missing values be handled before training?
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Exercise 3: Exploratory Data Analysis

Create:

- Histogram of MPG
- Box plot of MPG
- Correlation matrix
- Correlation bar chart
- Scatter plots:
 - MPG vs Weight
 - MPG vs Horsepower
 - MPG vs Cylinders
 - MPG vs Displacement
 - MPG vs Acceleration
 - MPG vs Model Year

For each scatter plot answer:

- Is the relationship positive or negative?

- Is it strong or weak?
 - Are there any outliers?
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